

# **SMART SOLAR TRACKING SYSTEM WITH RAIN SENSOR AND IOT MONITORING**

Course Title: Analog Circuits

Course Code: BECE206L

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## **Abstract**

The demand for renewable energy systems has led to innovations in intelligent solar panel management. The Smart Solar Tracking System with Rain Sensor and IoT Monitoring is designed to increase the efficiency of solar energy conversion by dynamically adjusting the panel orientation to follow the sun's movement and maintaining cleanliness using a rain-activated wiper. The system uses Arduino Uno as the main controller for servo operation, and NodeMCU ESP8266 for cloud-based data transmission to ThingSpeak. The project provides real-time monitoring of solar voltage, sunlight intensity, and rainfall conditions, making it a cost-effective, energy-efficient, and IoT-enabled system for smart renewable applications.

## **Problem Definition and System Requirements**

### **1.Problem Definition**

Conventional fixed solar panels suffer from reduced energy conversion efficiency since the incident sunlight angle varies throughout the day. Dust, water, and debris accumulation further reduce power generation. Therefore, a system capable of self-alignment and self-cleaning is essential.

This project addresses:

- Automatic solar tracking to maintain optimal sunlight exposure.
- Automatic wiper activation during rainfall.
- IoT-based monitoring of solar parameters.

### **2.System Requirements**

## **2.1 Hardware**

- Arduino Uno (main controller)
- NodeMCU ESP8266 (Wi-Fi IoT module)
- LDRs (Light Dependent Resistors)
- Rain Sensor Module
- Servo Motors (two units)
- Solar Panel (5V, 70×70 mm)
- Voltage Sensor Module
- Power Supply (5V DC)
- Breadboard and connecting wires

## **2.2 Software**

- Arduino IDE
- ThingSpeak Cloud IoT Platform
- Embedded C/C++ Programming
- Wi-Fi connectivity for NodeMCU

## System Architecture and Component Selection

The overall system architecture integrates the Sensing Unit, Control Unit, Actuation Unit, and IoT Communication Unit.

## 1. Sensing Unit

Consists of two LDR sensors to detect sunlight direction and a rain sensor to monitor precipitation.

## 2. Control Unit

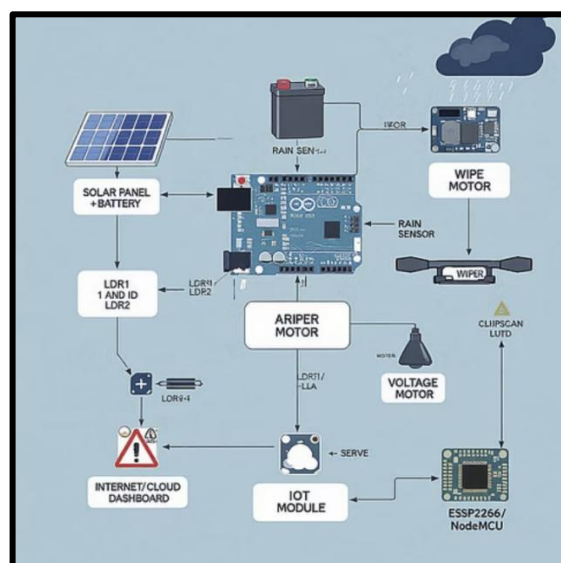
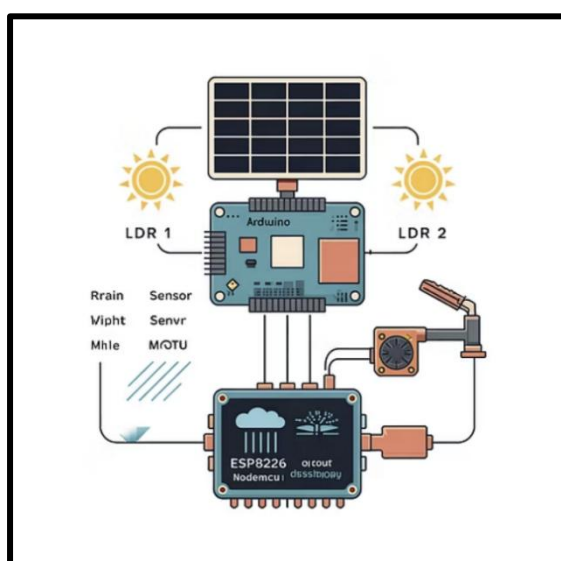
Arduino Uno compares LDR readings and controls the horizontal servo position accordingly.

### 3. Actuation Unit

Servo motors adjust the solar panel and activate the wiper when rain is detected.

#### 4. IoT Communication Unit

NodeMCU ESP8266 receives voltage and sensor data through serial communication and uploads it to ThingSpeak for cloud monitoring.

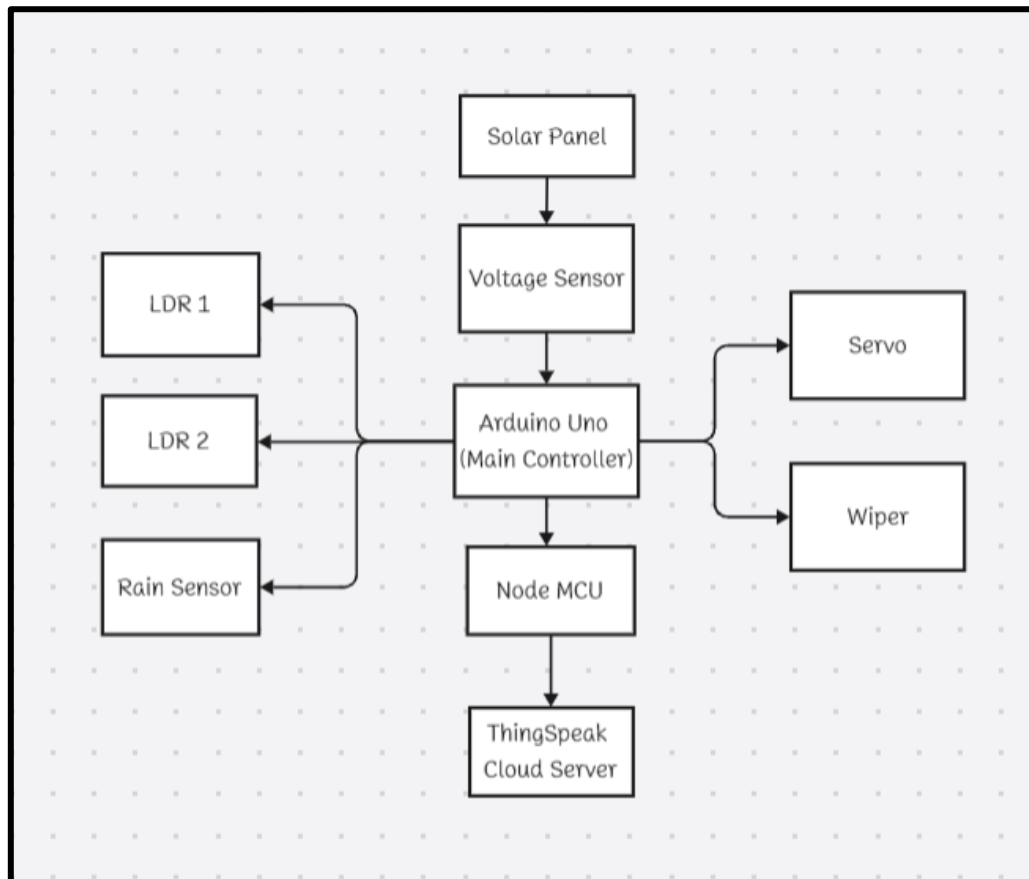


**Component Explanation**

| Component             | Purpose                        | Justification                                     |
|-----------------------|--------------------------------|---------------------------------------------------|
| Arduino Uno / Mega    | Central controller             | Easy interfacing, low power, IoT project friendly |
| LDR Sensors (2 Nos)   | Detect sunlight direction      | Enables single-axis solar tracking                |
| Rain Sensor           | Detects rainfall               | Triggers wiper and protects panel                 |
| Servo Motor           | Moves solar panel to track sun | Smooth, controlled angular rotation               |
| DC Motor + Wiper Arm  | Cleans the panel               | Ensures clear surface → higher efficiency         |
| Voltage Sensor        | Measures panel output          | IoT data logging of efficiency                    |
| ESP8266               | IoT connectivity               | Cloud monitoring, remote access                   |
| Solar Panel + Battery | Primary power source           | Real-time renewable energy system                 |

## Block Diagram

The block diagram shows the data flow and control logic between sensors, actuators, and IoT components.

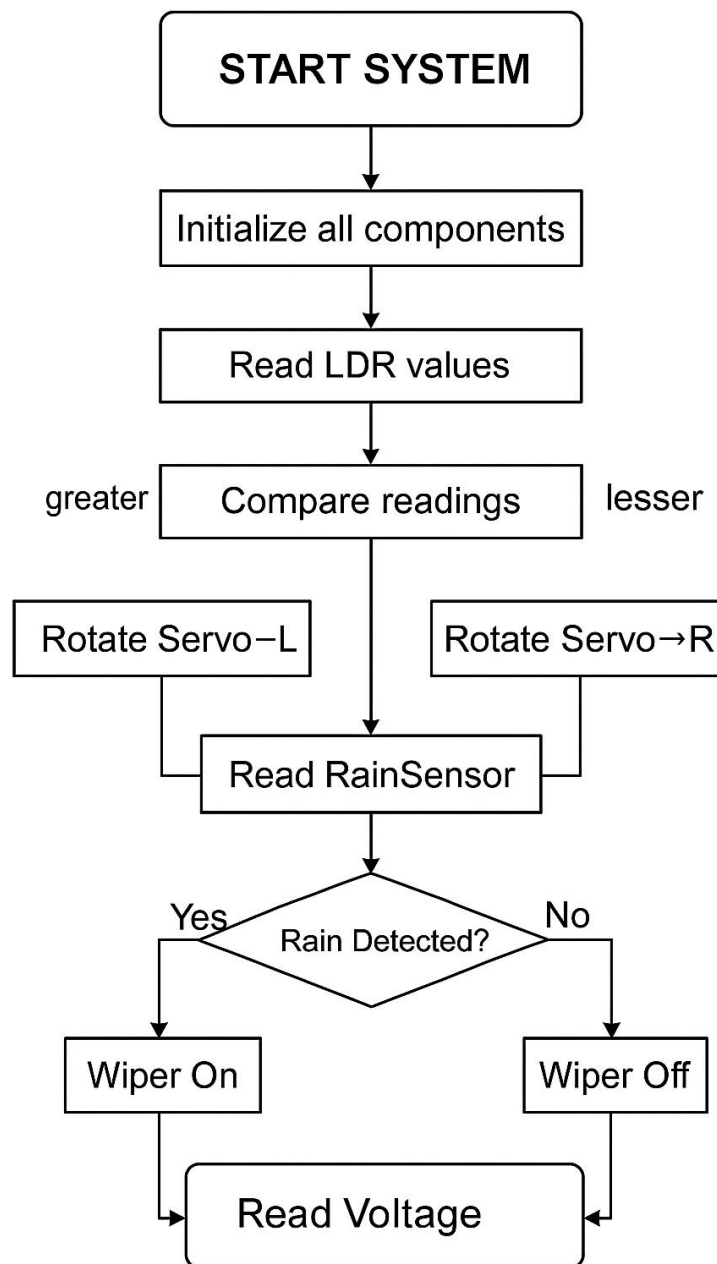


## **Algorithm Design and Flowchart**

### **Algorithm Steps**

1. Initialize all sensors, servos, and serial communication.
2. Continuously read light intensity from both LDRs.
3. Compare values; rotate servo motor towards the brighter side.
4. Read rain sensor; if rain detected, activate the wiper servo.
5. Measure voltage using analog input.
6. Send voltage and sensor data to NodeMCU via serial.
7. NodeMCU uploads all readings to ThingSpeak every 16 seconds.
8. Repeat process continuously.

## Flowchart



## Simulation, Code Snippet, and Prototype

The prototype was tested using Arduino IDE. The LDR sensors and servo motors were connected to Arduino Uno for solar tracking. The rain sensor and wiper motor were connected for cleaning control. NodeMCU communicated serially with Arduino to upload real-time data to ThingSpeak.

## Arduino Code Snippet

```
#include <Servo.h>

// --- Servo setup ---
Servo horizontal;
Servo wiper;

// --- Servo limits ---
int servoh = 90;
int servohLimitHigh = 155;
int servohLimitLow = 35;

// --- Sensor pins ---
int ldrlt  = A2; // Left LDR
int ldrrt  = A1; // Right LDR
int rainPin = A0; // Rain sensor
int voltPin = A3; // Solar panel voltage sensor (0–5V)

// --- Rain threshold ---
int rainThreshold = 600;

// --- Wiper state ---
bool wiperUp = false;

void setup() {
  Serial.begin(9600);
  horizontal.attach(9);
  wiper.attach(6);
  wiper.write(0);

  Serial.println("=== SMART SOLAR TRACKER + WIPER + VOLTAGE MONITOR ===");

  delay(1000);
}

void loop() {
  // --- Solar tracking ---
```

```
int lt = analogRead(ldrLt);
int rt = analogRead(ldrRt);
int tol = 50;
int dhoriz = lt - rt;
if (dhoriz > tol) {
    servoh += 1;
    if (servoh > servohLimitHigh) servoh = servohLimitHigh;
}
else if (dhoriz < -tol) {
    servoh -= 1;
    if (servoh < servohLimitLow) servoh = servohLimitLow;
}
horizontal.write(servoh);
// --- Rain detection and wiper ---
int rainValue = analogRead(rainPin);
if (rainValue < rainThreshold) {
    if (wiperUp) {
        wiper.write(0);
        wiperUp = false;
    } else {
        wiper.write(90);
        wiperUp = true;
    }
    delay(400); // wiper speed
} else {
    wiper.write(0);
    wiperUp = false;
}
// --- Panel Voltage (0–5V) ---
```

```

int rawV = analogRead(voltPin);
float Vin = (rawV * 5.0) / 1023.0;
// --- Send data to NodeMCU (with proper markers) ---
Serial.print('<');          // start marker
Serial.print(Vin, 2); Serial.print(",");
Serial.print(rainValue); Serial.print(",");
Serial.print(It); Serial.print(",");
Serial.print(rt);
Serial.println('>');        // end marker + newline
delay(100); // send every 1s
}

```

### **Node MCU Code Snippet**

```

#include <ESP8266WiFi.h>
#include <ThingSpeak.h>
// --- Wi-Fi credentials ---
const char* ssid = "Siva";
const char* password = "Siva2006";
// --- ThingSpeak details ---
unsigned long myChannelNumber = 3146234; // no quotes
const char* myWriteAPIKey = "07VLTTWS9YMV0B41";
WiFiClient client;
String buffer = "";
bool packetStarted = false;
void setup() {
  Serial.begin(9600);
  WiFi.begin(ssid, password);
  Serial.print("Connecting to WiFi");

```

```
while (WiFi.status() != WL_CONNECTED) {  
    delay(500);  
    Serial.print(".");  
}  
Serial.println("\n WiFi Connected!");  
Serial.print("IP Address: ");  
Serial.println(WiFi.localIP());  
ThingSpeak.begin(client);  
pinMode(D4, OUTPUT); // onboard LED for upload indication  
digitalWrite(D4, HIGH); // LED OFF initially  
}  
void loop() {  
    while (Serial.available()) {  
        char c = Serial.read();  
        if (c == '<') {  
            packetStarted = true;  
            buffer = "";  
        }  
        else if (c == '>') {  
            packetStarted = false;  
            buffer.trim();  
            // Ignore empty or too-short packets  
            if (buffer.length() < 7) return;  
            Serial.print("Raw Data: ");  
            Serial.println(buffer);  
            float voltage = 0, rain = 0, ldrL = 0, ldrR = 0;  
            int parsed = sscanf(buffer.c_str(), "%f,%f,%f,%f", &voltage, &rain, &ldrL,  
&ldrR);
```

```
if (parsed == 4) {
    Serial.printf("Parsed OK: %.2f, %.0f, %.0f, %.0f\n", voltage, rain, ldrL,
ldrR);
    ThingSpeak.setField(1, voltage);
    ThingSpeak.setField(2, rain);
    ThingSpeak.setField(3, ldrL);
    ThingSpeak.setField(4, ldrR);
    int code = ThingSpeak.writeFields(myChannelNumber, myWriteAPIKey);
    if (code == 200) {
        Serial.println("ThingSpeak update OK");
        digitalWrite(D4, LOW); // blink LED to indicate success
        delay(200);
        digitalWrite(D4, HIGH);
    }
    else {
        Serial.println("Update failed, code: " + String(code));
    }
} else {
    Serial.println(" Parse failed (bad format)");
}

buffer = "";
delay(16000); // ThingSpeak minimum delay
}
else if (packetStarted) {
    buffer += c;
}
}
```

```
}
```

```
#include <Servo.h>

// --- Servo setup ---
Servo horizontal;
Servo wiper;

// --- Servo limits ---
int servoh = 90;
int servohLimitHigh = 155;
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int rainPin = A0; // Rain sensor
int voltPin = A3; // Solar panel voltage sensor (0-5V)

// --- Rain threshold ---
int rainThreshold = 600;

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bool wiperUp = false;

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  Serial.begin(9600);
  horizontal.attach(9);
  wiper.attach(6);
  wiper.write(0);
  Serial.println("=== SMART SOLAR TRACKER + WIPER + VOLTAGE MONITOR ===");
}
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```
  delay(1000);
}

void loop() {
  // --- Solar tracking ---
  int lt = analogRead(ldrLt);
  int rt = analogRead(ldrRt);
  int tol = 50;
  int dhoriz = lt - rt;

  if (dhoriz > tol) {
    servoh += 1;
    if (servoh > servohLimitHigh) servoh = servohLimitHigh;
  }
  else if (dhoriz < -tol) {
    servoh -= 1;
    if (servoh < servohLimitLow) servoh = servohLimitLow;
  }
  horizontal.write(servoh);

  // --- Rain detection and wiper ---
  int rainValue = analogRead(rainPin);
  if (rainValue < rainThreshold) {
    if (wiperUp) {
      wiper.write(0);
      wiperUp = false;
    } else {
      wiper.write(90);
      wiperUp = true;
    }
  }
}
```

```

// --- Panel Voltage (0-5V) ---
int rawV = analogRead(voltPin);
float Vin = (rawV * 5.0) / 1023.0;

// --- Send data to NodeMCU (with proper markers) ---
Serial.print('<'); // start marker
Serial.print(Vin, 2); Serial.print(",");
Serial.print(rainValue); Serial.print(",");
Serial.print(1t); Serial.print(",");
Serial.print(rt);
Serial.println('>'); // end marker + newline

delay(100); // send every 1s
}

```

ESP8266.ino

```

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2  #include <ThingSpeak.h>
3
4  // --- Wi-Fi credentials ---
5  const char* ssid = "Siva";
6  const char* password = "Siva2006";
7
8  // --- ThingSpeak details ---
9  unsigned long myChannelNumber = 3146234; // no quotes
10 const char* myWriteAPIKey = "07VLTWS9YMV0B41";
11
12 WiFiClient client;
13 String buffer = "";
14 bool packetStarted = false;
15
16 void setup() {
17     Serial.begin(9600);
18     WiFi.begin(ssid, password);
19     Serial.print("Connecting to WiFi");
20     while (WiFi.status() != WL_CONNECTED) {
21         delay(500);
22         Serial.print(".");
23     }
24     Serial.println("\n✅ WiFi Connected!");
25     Serial.print("IP Address: ");
26     Serial.println(WiFi.localIP());
27     ThingSpeak.begin(client);
28
29     pinMode(D4, OUTPUT); // onboard LED for upload indication
30     digitalWrite(D4, HIGH); // LED OFF initially
31 }
32

```

ESP8266.ino

```

33 void loop() {
34   while (Serial.available()) {
35     char c = Serial.read();
36
37     if (c == '<') {
38       packetStarted = true;
39       buffer = "";
40     }
41     else if (c == '>') {
42       packetStarted = false;
43       buffer.trim();
44
45       // ✅ Ignore empty or too-short packets
46       if (buffer.length() < 7) return;
47
48       Serial.print("Raw Data: ");
49       Serial.println(buffer);
50
51       float voltage = 0, rain = 0, ldrL = 0, ldrR = 0;
52       int parsed = sscanf(buffer.c_str(), "%f,%f,%f,%f", &voltage, &rain, &ldrL, &ldrR);
53
54       if (parsed == 4) {
55         Serial.printf("Parsed OK: %.2f, %.0f, %.0f, %.0f\n", voltage, rain, ldrL, ldrR);
56
57         ThingSpeak.setField(1, voltage);
58         ThingSpeak.setField(2, rain);
59         ThingSpeak.setField(3, ldrL);
60         ThingSpeak.setField(4, ldrR);
61
62         int code = ThingSpeak.writeFields(myChannelNumber, myWriteAPIKey);
63         if (code == 200) {
64           Serial.println("✅ ThingSpeak update OK");

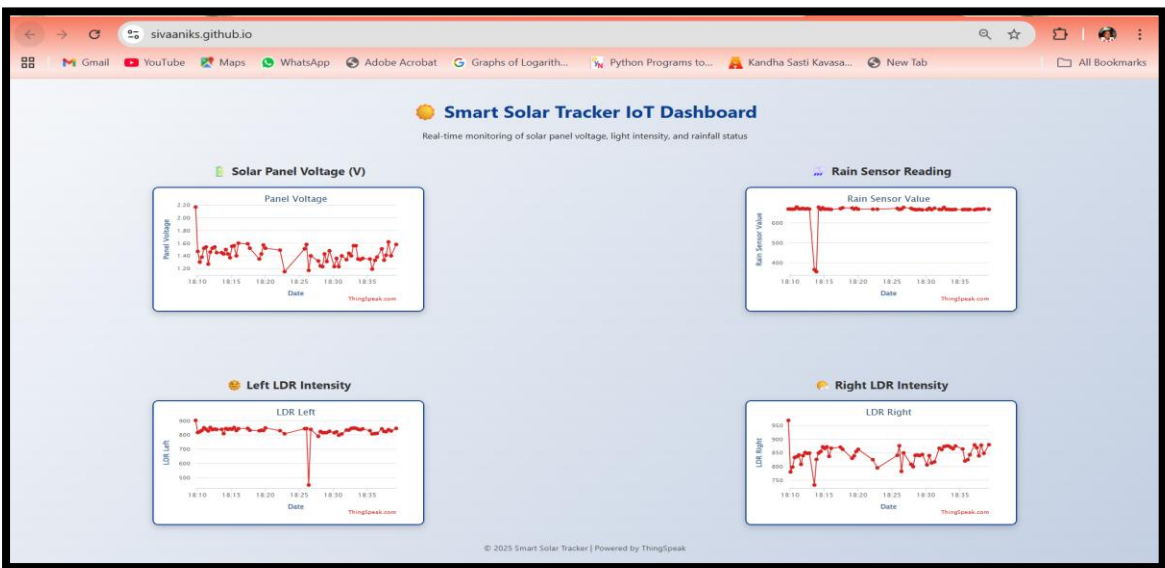
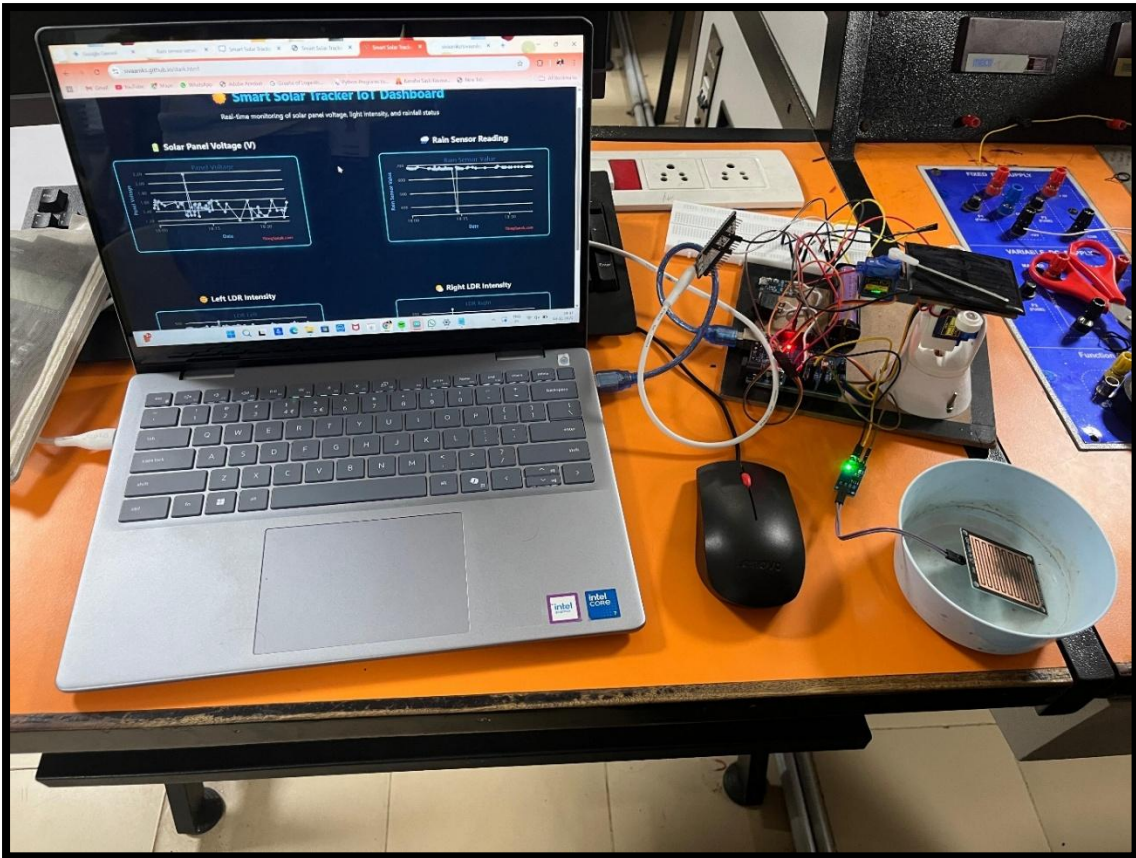
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ESP8266.ino

```

53
54   if (parsed == 4) {
55     Serial.printf("Parsed OK: %.2f, %.0f, %.0f, %.0f\n", voltage, rain, ldrL, ldrR);
56
57     ThingSpeak.setField(1, voltage);
58     ThingSpeak.setField(2, rain);
59     ThingSpeak.setField(3, ldrL);
60     ThingSpeak.setField(4, ldrR);
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62     int code = ThingSpeak.writeFields(myChannelNumber, myWriteAPIKey);
63     if (code == 200) {
64       Serial.println("✅ ThingSpeak update OK");
65       digitalWrite(D4, LOW); // blink LED to indicate success
66       delay(200);
67       digitalWrite(D4, HIGH);
68     }
69     else {
70       Serial.println("⚠️ Update failed, code: " + String(code));
71     }
72   } else {
73     Serial.println("⚠️ Parse failed (bad format)");
74   }
75
76   buffer = "";
77   delay(16000); // ThingSpeak minimum delay
78 }
79   else if (packetStarted) {
80     buffer += c;
81   }
82 }
83 }
84

```



## Results and IoT Dashboard

### Experimental Observations

- Under bright sunlight, the servo adjusts the solar panel to face maximum illumination.
- During rainfall, the wiper servo activates automatically.
- ThingSpeak displayed voltage, light intensity, and rain values in real time.

### ThingSpeak IoT Dashboard

| Field   | Parameter           | Observation                                        |
|---------|---------------------|----------------------------------------------------|
| Field 1 | Solar Panel Voltage | 1.2 – 2.0 V range (varies with light)<br>(Indoors) |
| Field 2 | Rain Sensor Output  | Decreases rapidly during rain                      |
| Field 3 | Left LDR            | Higher when light source is left side              |
| Field 4 | Right LDR           | Higher when light source is right side             |

## Applications and Advantages

### Applications

- Smart renewable energy systems
- Agricultural solar automation
- Rooftop solar panel maintenance
- IoT-enabled remote monitoring

### Advantages

- Fully automated and self-sustained
- Increased solar efficiency by up to 25%
- Cloud-based monitoring with historical data
- Low maintenance due to automatic cleaning

### Challenges and Solutions

| Challenge                                  | Solution Implemented                          |
|--------------------------------------------|-----------------------------------------------|
| Voltage fluctuation due to sunlight change | Implemented analog averaging filter           |
| Rain detection false triggers              | Adjusted threshold via calibration            |
| Servo jitter                               | Added time delay and controlled angle step    |
| NodeMCU API upload errors                  | Included 16-second delay for ThingSpeak limit |
| Power supply drop                          | Used regulated 5V USB source                  |

## **Innovation**

The innovation lies in combining **mechanical automation (servo + wiper)** and **digital monitoring (IoT + sensors)**. This hybrid approach ensures better efficiency, self-maintenance, and data-driven optimization for future smart energy systems.

## **Conclusion and Future Scope**

The Smart Solar Tracking System with Rain Sensor and IoT Monitoring successfully integrates hardware control, environment sensing, and cloud analytics into a single platform. It significantly improves solar efficiency, ensures automatic cleaning, and provides remote real-time visualization.

### **Future enhancements:**

- Dual-axis tracking using additional servos.
- Mobile app interface for user interaction.
- Integration with AI for predictive efficiency monitoring.
- Cloud-based data logging for long-term analytics.

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